

Background



- The UWAFST team wants to mount two LiDAR sensors on the front and back
- This would enable CAV to use the LiDAR for low-speed parking scenarios
- UWAFST goals is to implement these two LiDAR sensors with keeping functionality and aesthetics in mind

Design Constraints

1) Mount Angle $\leq 10^\circ$

2) Field of View $\geq 120^\circ$

3) Modular Design

4) Must fit in 3" Envelope

Design Criteria

1) Weatherproofing

2) Aesthetics

3) Structural Rigidity

4) Cost Effectiveness

Timeline

Task ID	Task Title	Start Date	Due Date	Pct of Task Complete	September				October				November				December			
					W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
1	LIDAR Placement		2025-10-06																	
1.1	Positional Research / Background Research	09-22-25	09-26-25	100%																
1.2	Define Design Requirements	09-22-25	09-26-25	100%																
1.3	Location Mapping / Prelim CAD Dimensions	10-01-25	10-06-25	100%																
2.0	Prototyping																			
2.1	Design																			
2.2	Software Learning (CAD, Siemens NX)	09-25-25	10-28-25	100%																
2.3	Mount Design - Front	10-06-25	10-28-25	100%																
2.4	Mount Design - Rear	10-06-25	10-28-25	100%																
2.7	Simulations																			
2.8	Vibrations	10-29-25	11-03-25	100%																
2.9	FEA	10-30-25	11-04-25	100%																
2.11	Vehicle Setup For Mounting																			
2.12	Protoyping	11-05-25	11-12-25	100%																
2.13	Integration																			
2.14	Electrical Routing	11-05-25	11-12-25	0%																
3	Aestheics																			
3.1	Material Selection	10-27-25	11-03-25	100%																
3.2	Surface Modeling	11-03-25	11-10-25	100%																
3.3	Manufacturing Methods	11-10-25	11-17-25	100%																
4.0	Testing and Calibration																			
4.1	LIDAR Calibration/ Validation	11-17-25	11-24-25	0%																

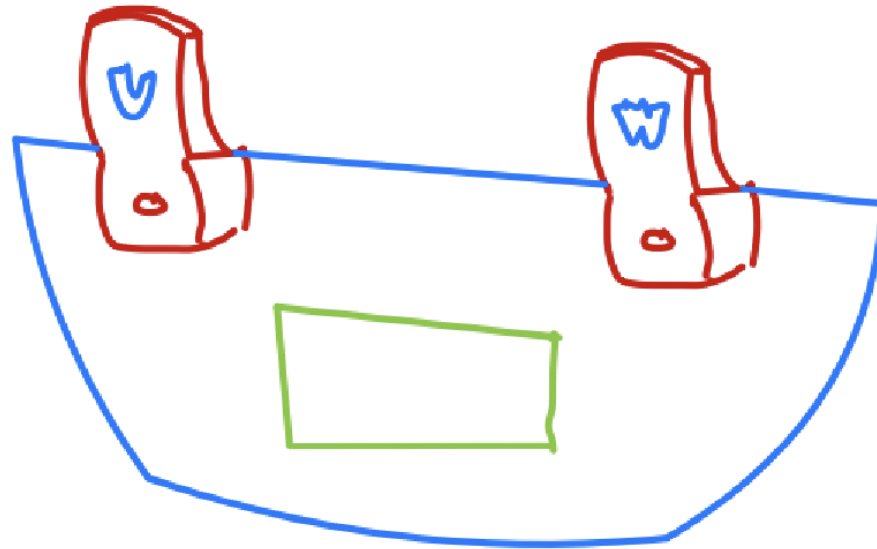
Engineering Methodology

Design Ideation

Concept Sketches

Sketch #1

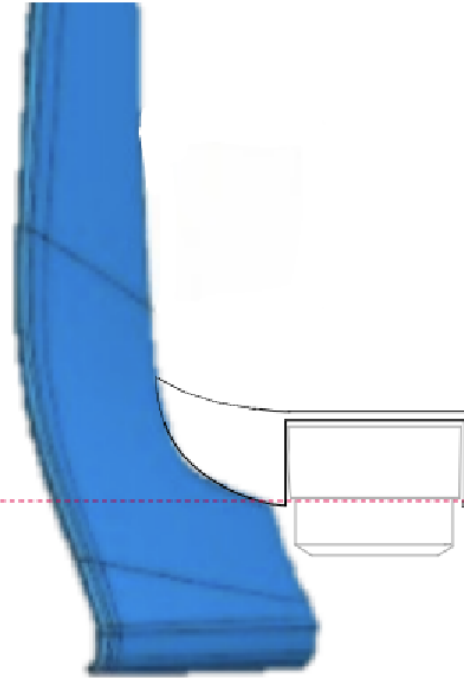
Part



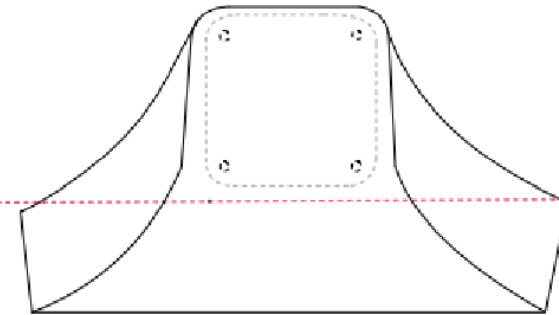
- under the trunk handle
- focus on curvature
- lidar → plate → hook mount
- focus on material selection

Sketch #2

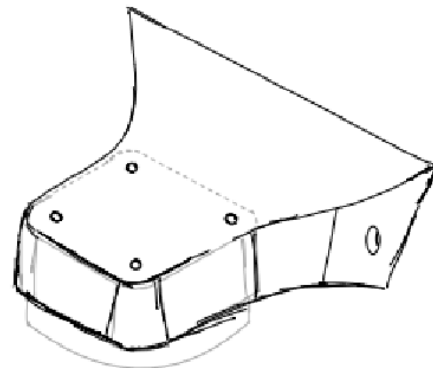
Side View



Top View

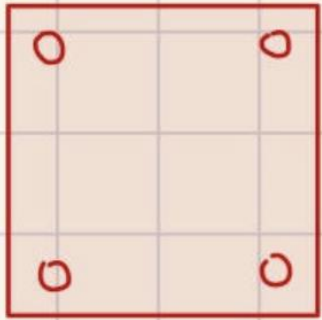


Isometric View

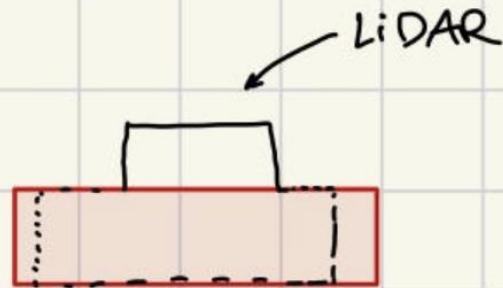


Sketch #3

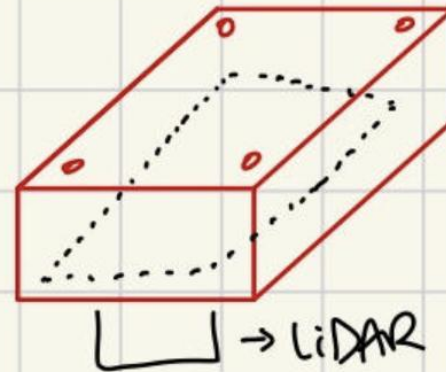
TOP VIEW



FRONT VIEW

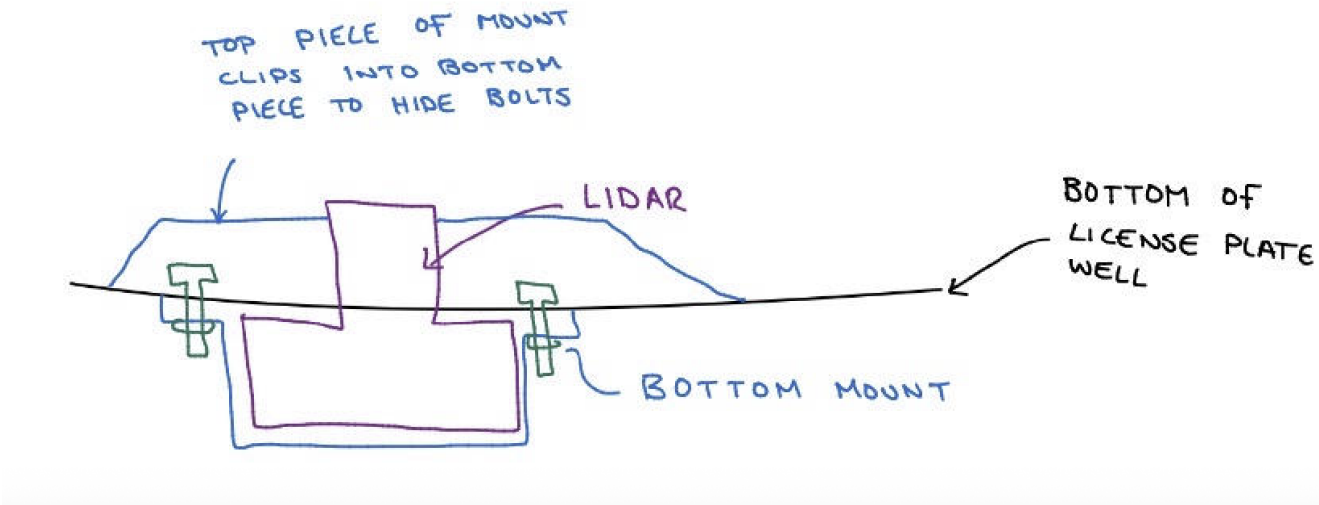
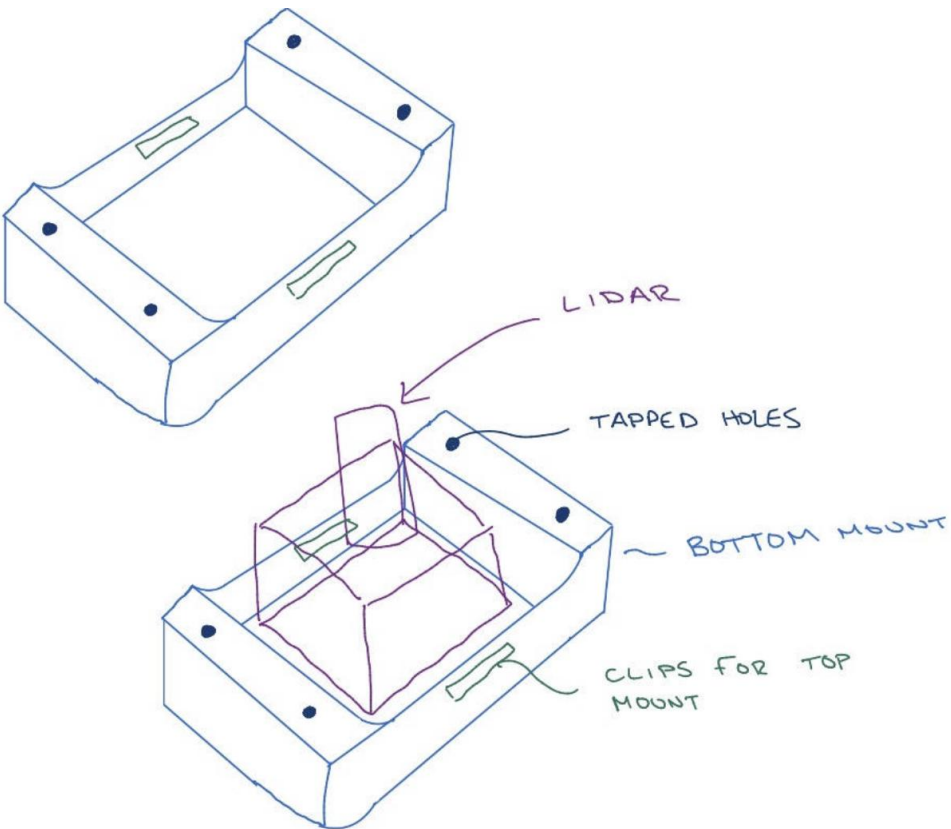


ISOMETRIC



- mounted upside down
- LiDAR sticking out so that it doesn't restrict fov
- cut into trunk (if possible) to make it flush.

Sketch #4



Decision Matrix

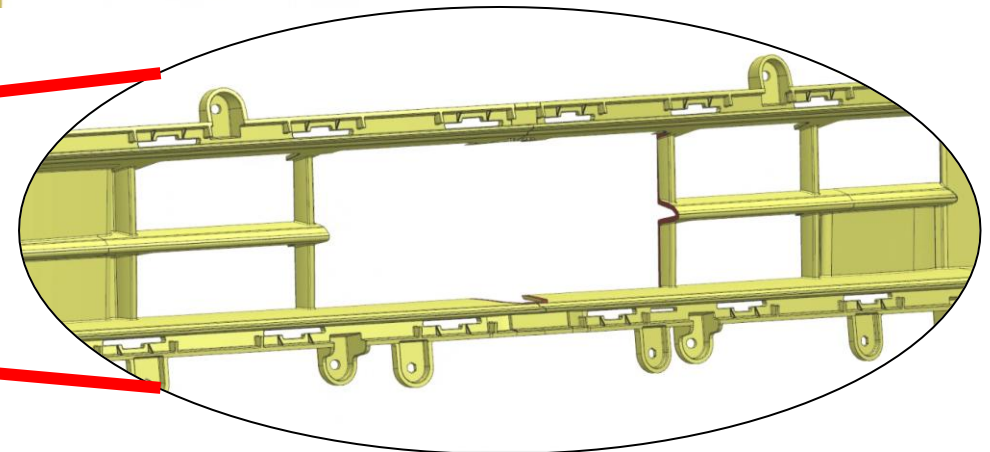
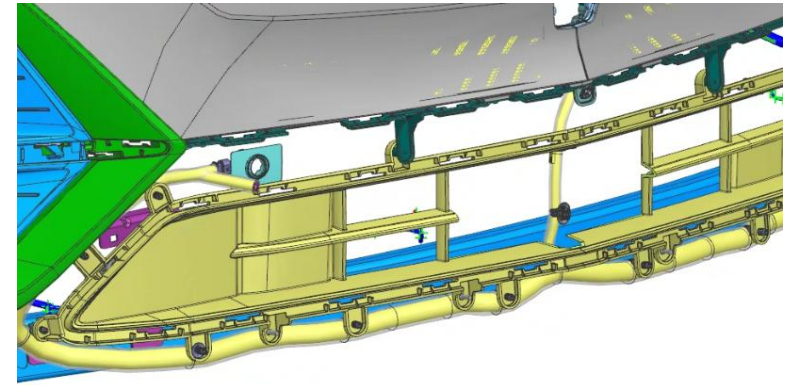
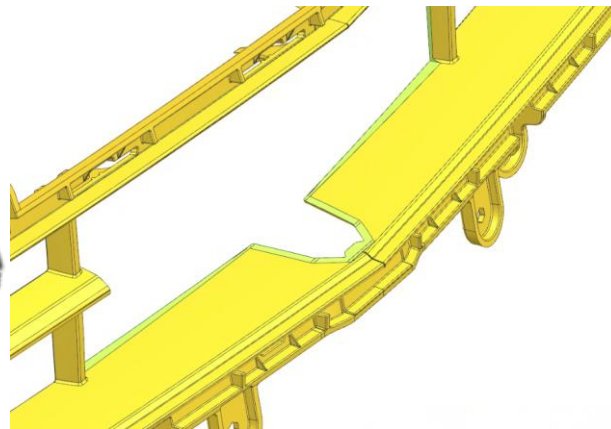
Design Criteria	Sketch #1	Sketch #2	Sketch #3	Sketch #4
Manufacturability	2	3	1	3
Durability	2	3	2	3
Aesthetics and integration with existing vehicle	3	3	1	4
Sensing and FOV	3	4	3	2
Serviceability and ease of access	2	3	2	1
Total Score	12	16	9	13

Design Modifications

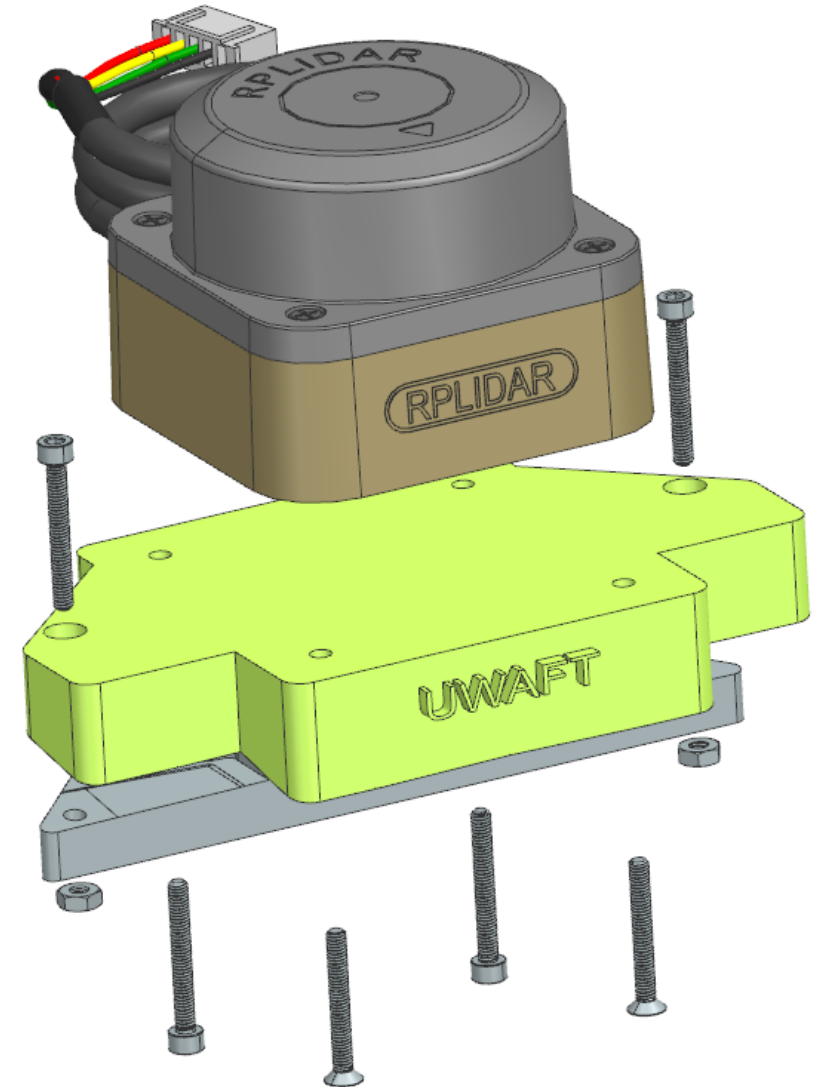
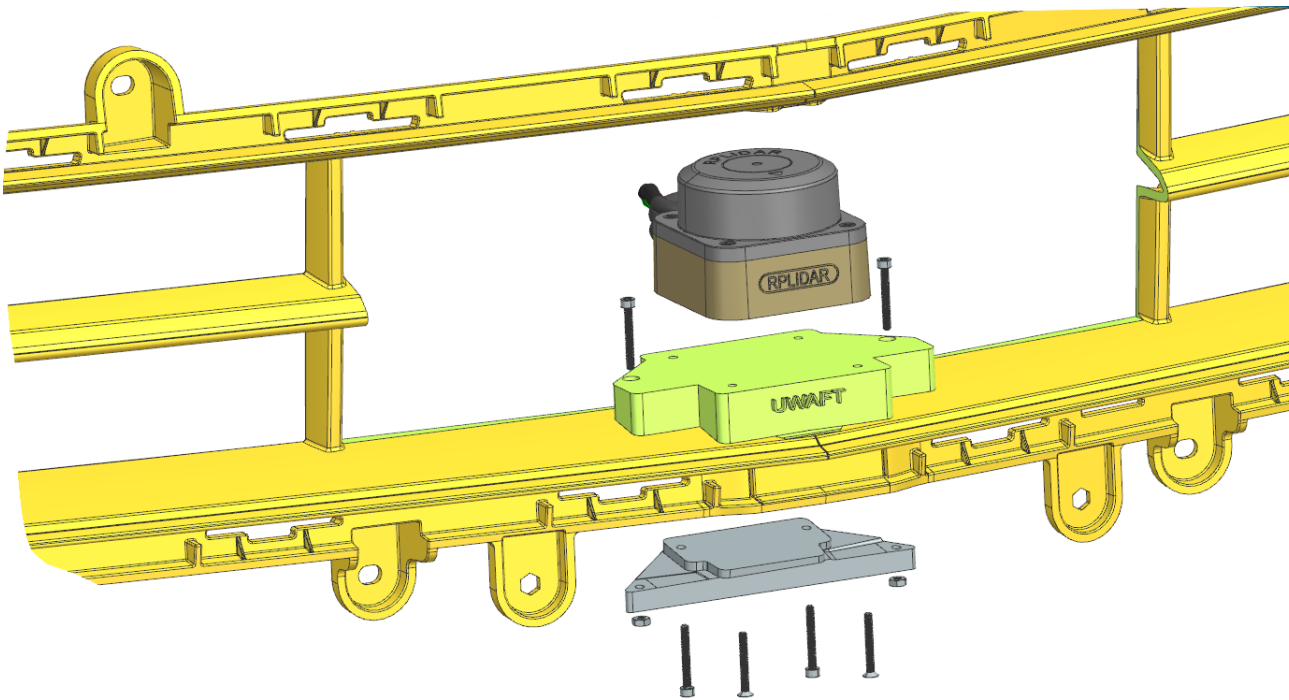
Front & Rear Mount

Front Mount Design

Front Mounting Location: Inside the lower fascia air inlet
(center plastic section is removed for LiDAR clearance)



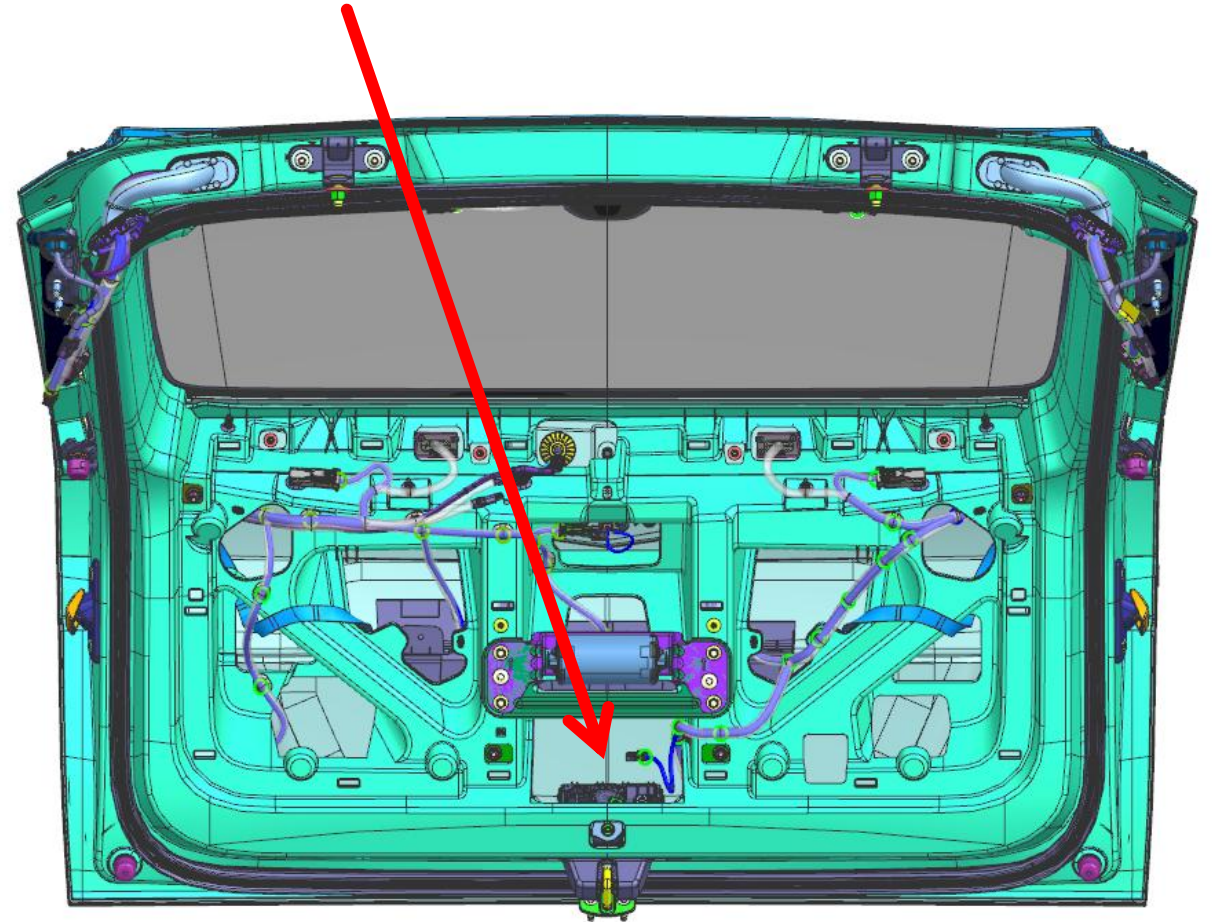
New Front Mount CAD



- LiDAR is mounted parallel to the ground plane
- Mount screws are accessible

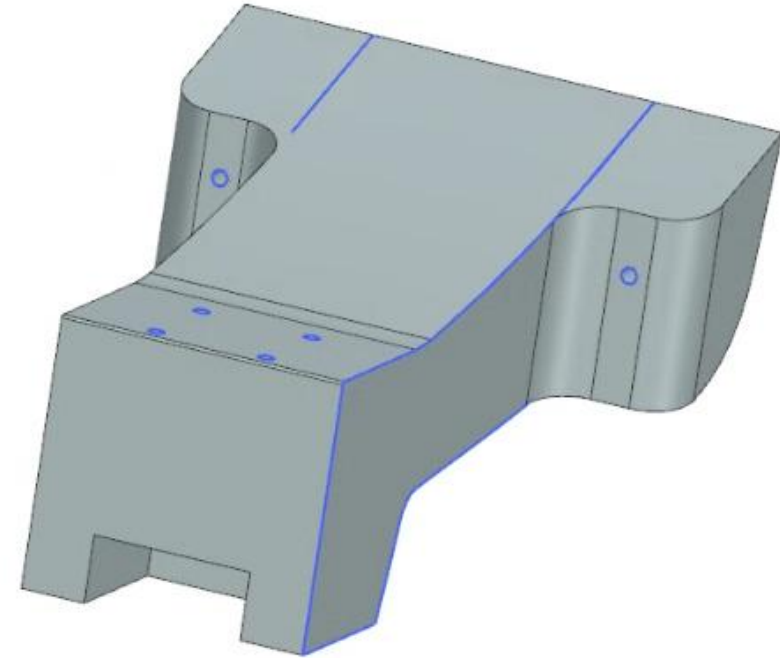
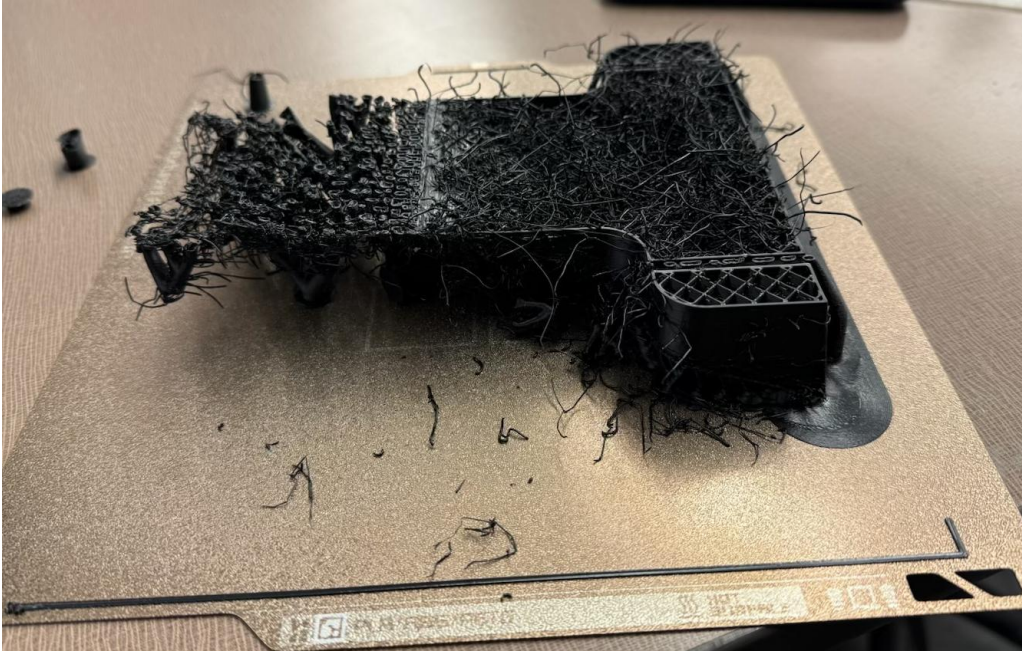
Rear Mount Design

Rear Mounting Design: Underneath the license plate bolting into the trunk



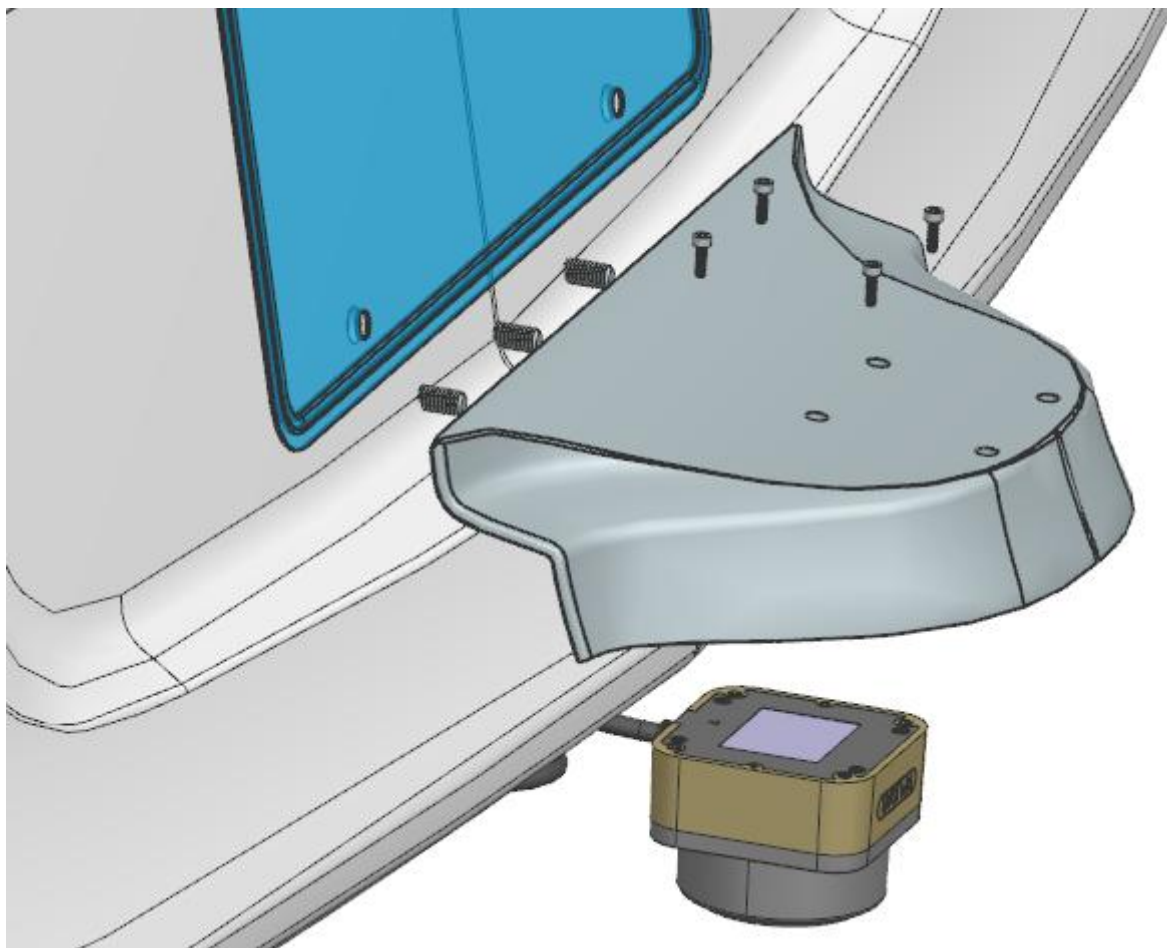
Obstacles & Challenges

1st Revision: Rear Mount Design



- First Iteration of Rear Mount Design failed printing multiple times
- Difficult to print and print time was over 25+ hours

New Rear Mount CAD



ANSYS Simulations

Design Validation

Design Validation

Simulation Parameters

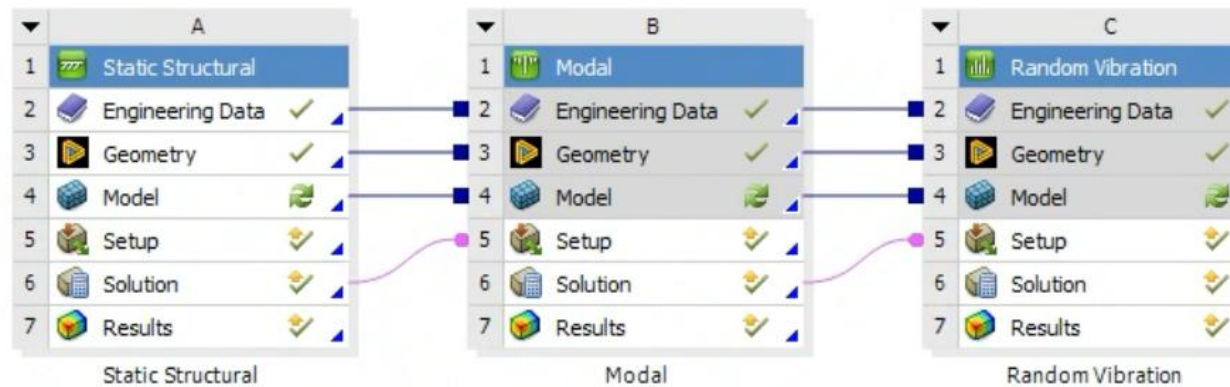


Figure 12: Example for Cross-Axis Acceleration Upper Limit for Random Vibration - Sprung Mass

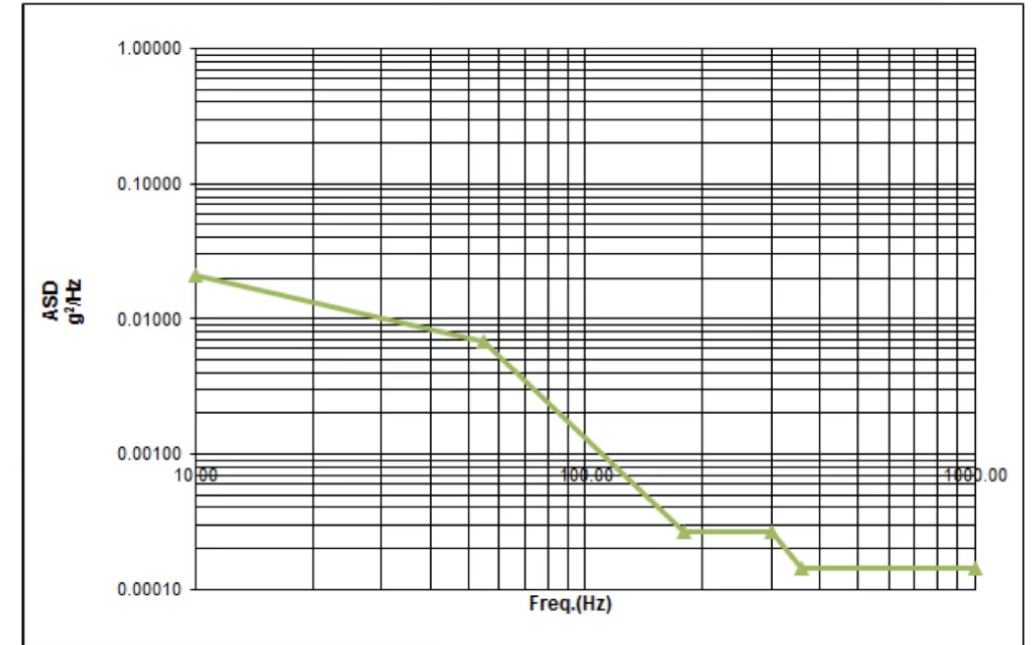


Table 18: Random Vibration Cross-Axis Acceleration Upper Limit for Sprung Mass

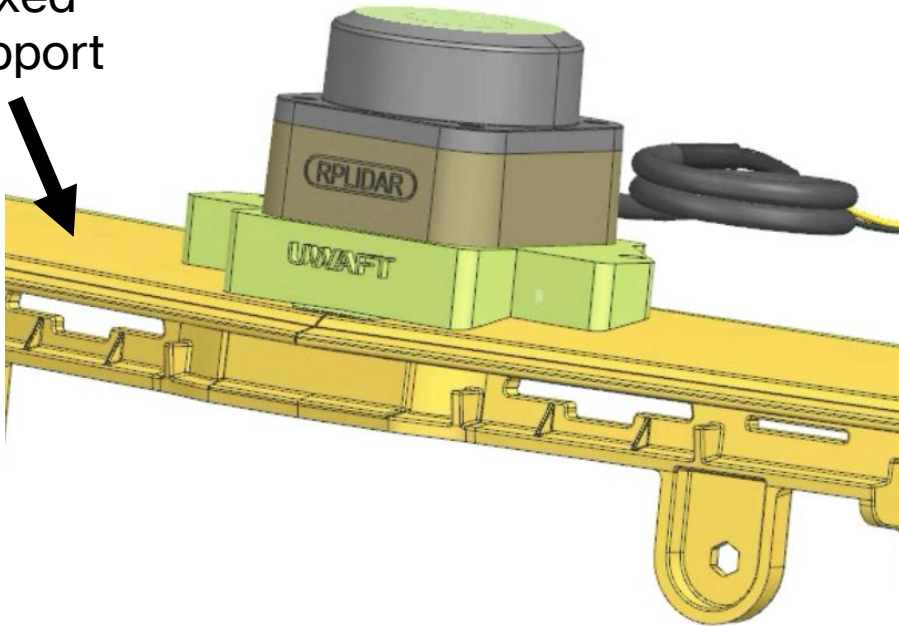
FREQ in Hz	PSD in $(m/s^2)^2/Hz$	PSD in g^2/Hz
10	2.0098	0.0209
55	0.6543	0.0068
180	0.0253	0.0003
300	0.0253	0.0003
360	0.0136	0.0001
1000	0.0136	0.0001

Effective Acceleration = $8.8 m/s^2 = 0.9 G_{RMS}$

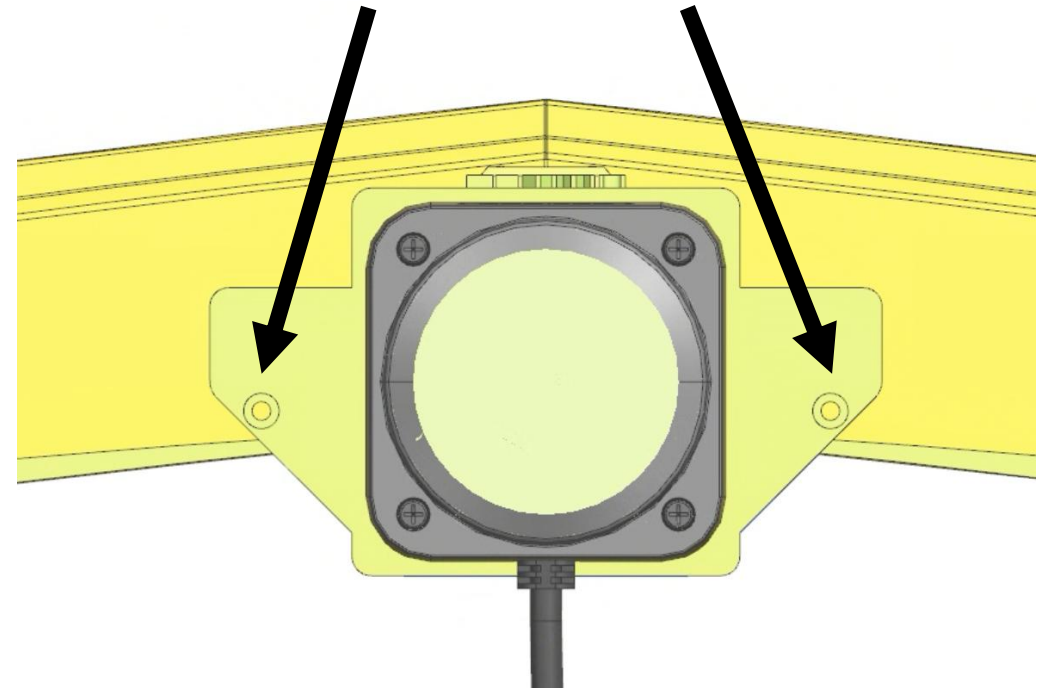
Front Mount Design Validation

Model Setup

Fixed
Support



Bolted Joint
Connection
(Preload Applied)

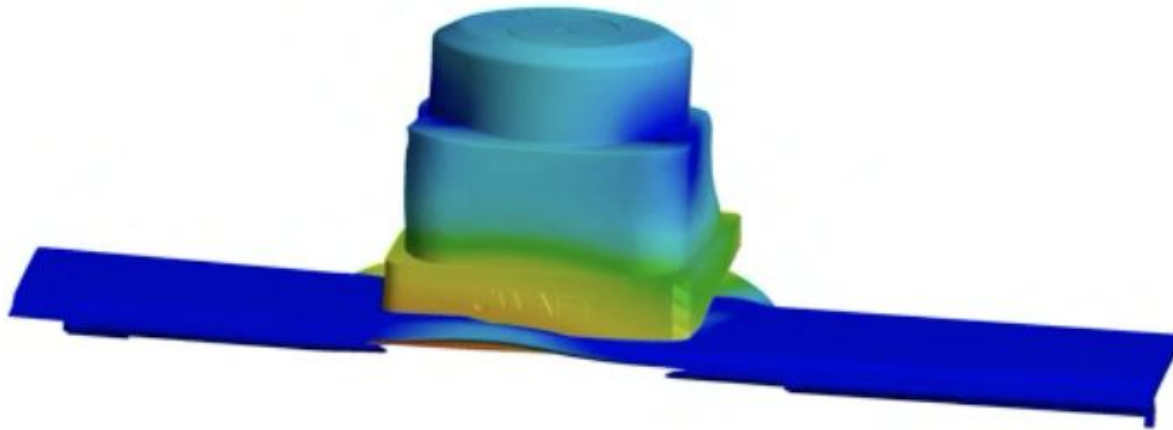
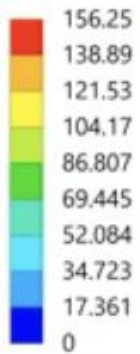


Front Mount Design Validation

Modal Results

Modal

Total Deformation 3
Type: Total Deformation
Frequency: 2051.7 Hz
Unit: mm
Max: 156.25
Min: 0
11/29/2025 9:31 PM



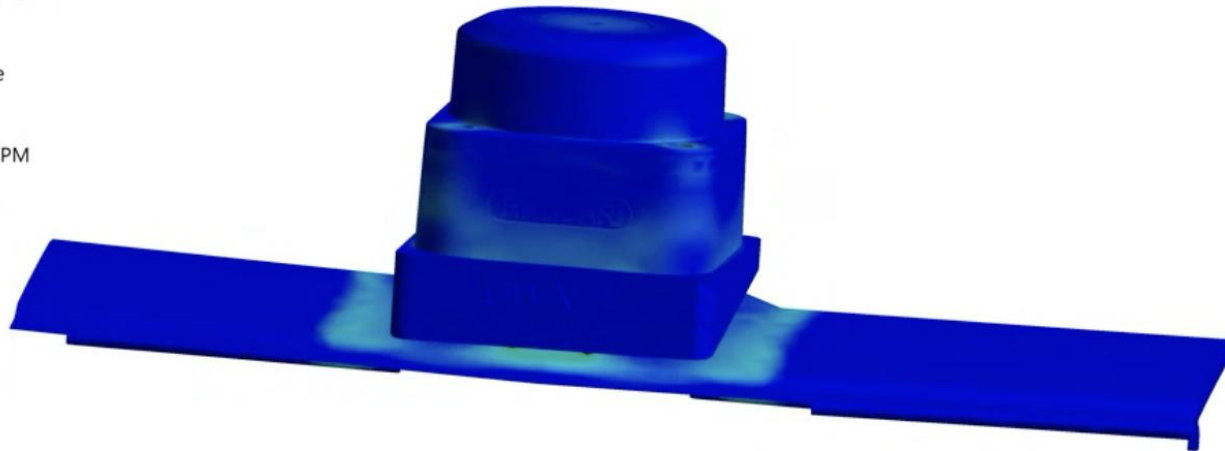
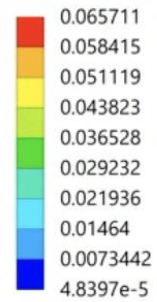
	Mode	☒ Frequency [Hz]
1	1.	1305.4
2	2.	1566.3
3	3.	2051.7
4	4.	2436.8
5	5.	2630.
6	6.	2799.2

Front Mount Design Validation

Random Vibration: Stresses

Random Vibration

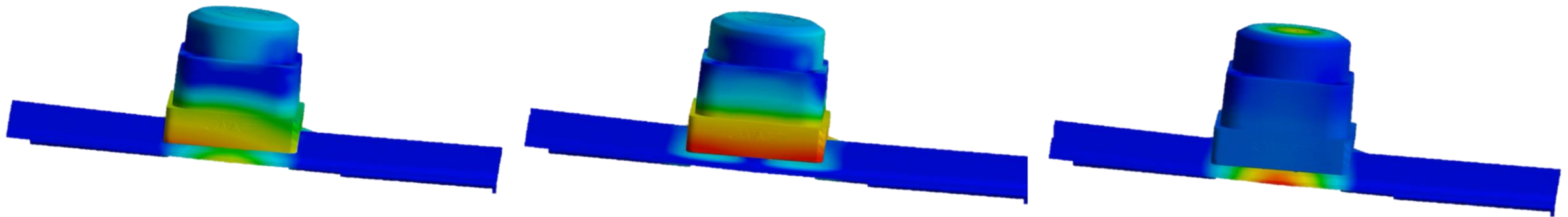
Equivalent Stress
Type: Equivalent Stress
Scale Factor Value: 3 Sigma
Probability: 99.73 %
Unit: MPa
Time: 0 s
Custom Obsolete
Max: 0.14827
Min: 6.7373e-8
11/30/2025 8:27 PM



**Bolt mounting
to Bumper**

Front Mount Design Validation

Random Vibration: Directional Deformation



X-Axis

Y-Axis

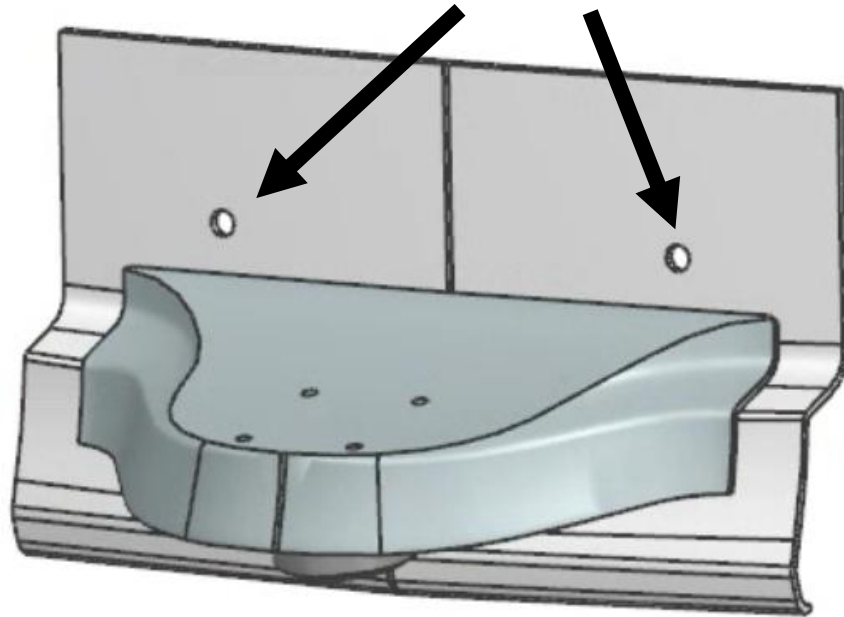
Z-Axis



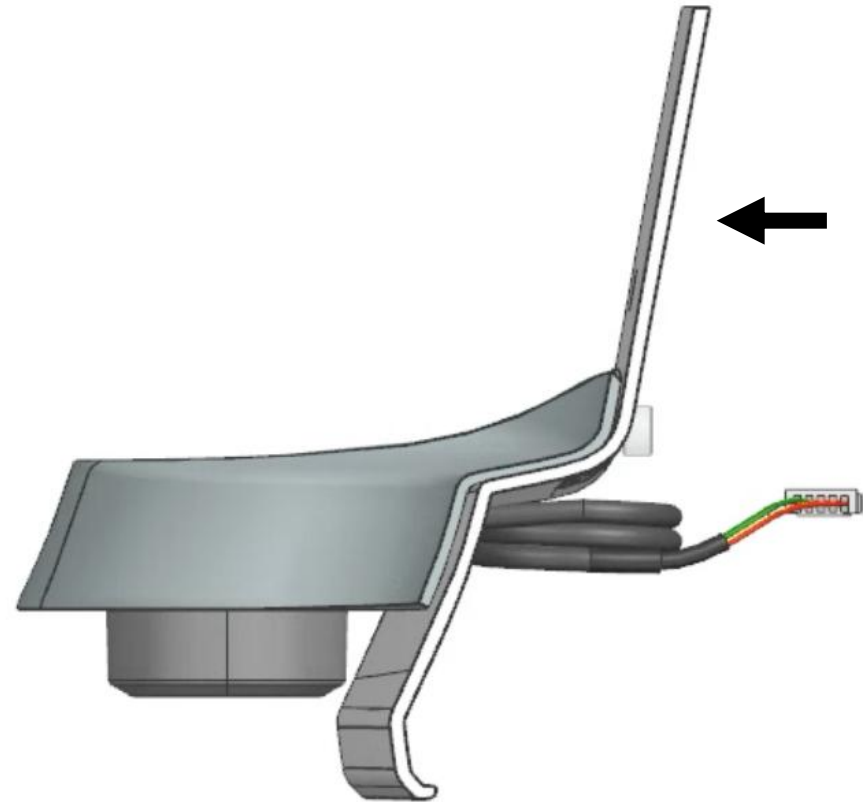
Rear Mount Design Validation

Model Setup

**Bolted Joint
Connection
(Preload Applied)**



**Fixed
Support**



Rear Mount Design Validation

Modal Results

B: Modal

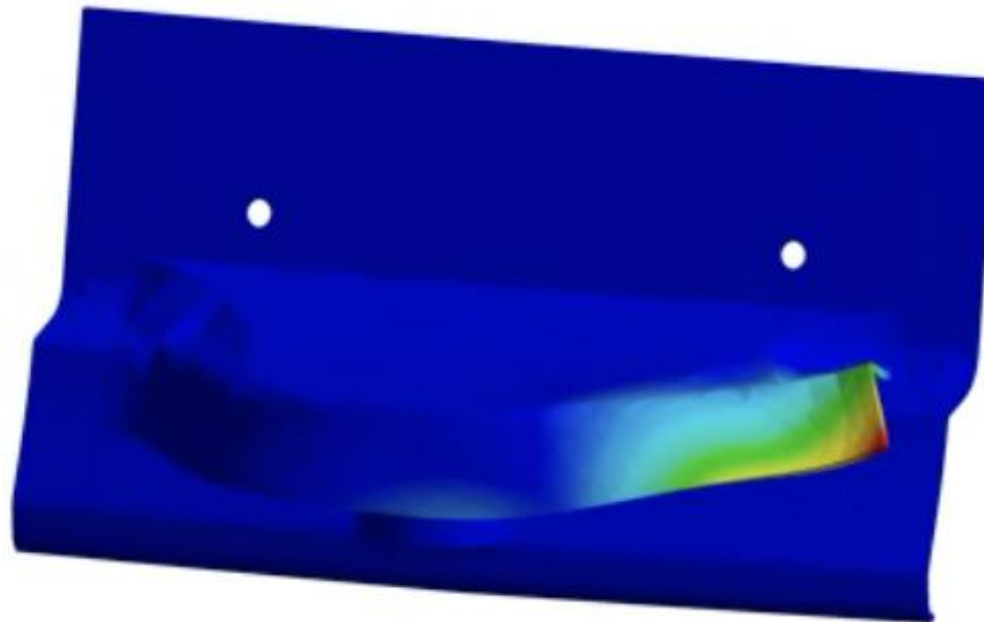
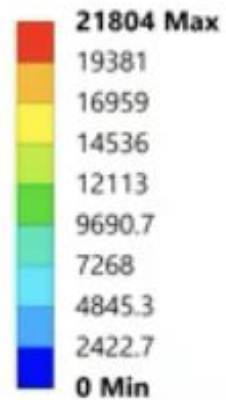
Total Deformation

Type: Total Deformation

Frequency: 575.05 Hz

Unit: mm

2025-11-30 7:56:21 PM

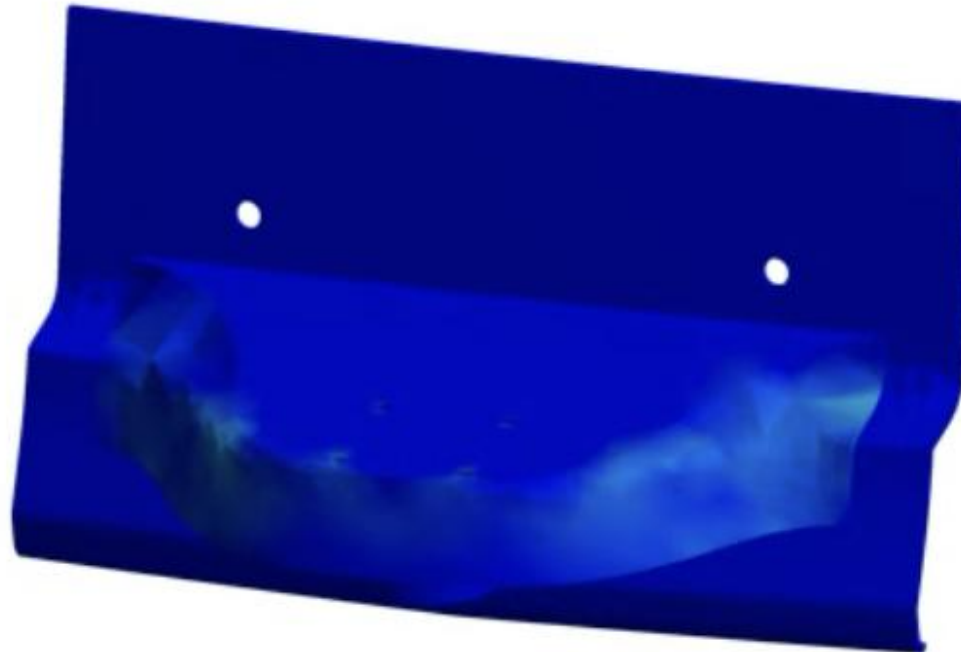
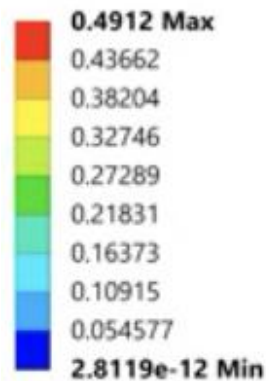


	Mode	☒ Frequency [Hz]
1	1.	575.05
2	2.	656.88
3	3.	805.56
4	4.	959.17
5	5.	997.73
6	6.	1071.9

Rear Mount Design Validation

Random Vibration: Stresses

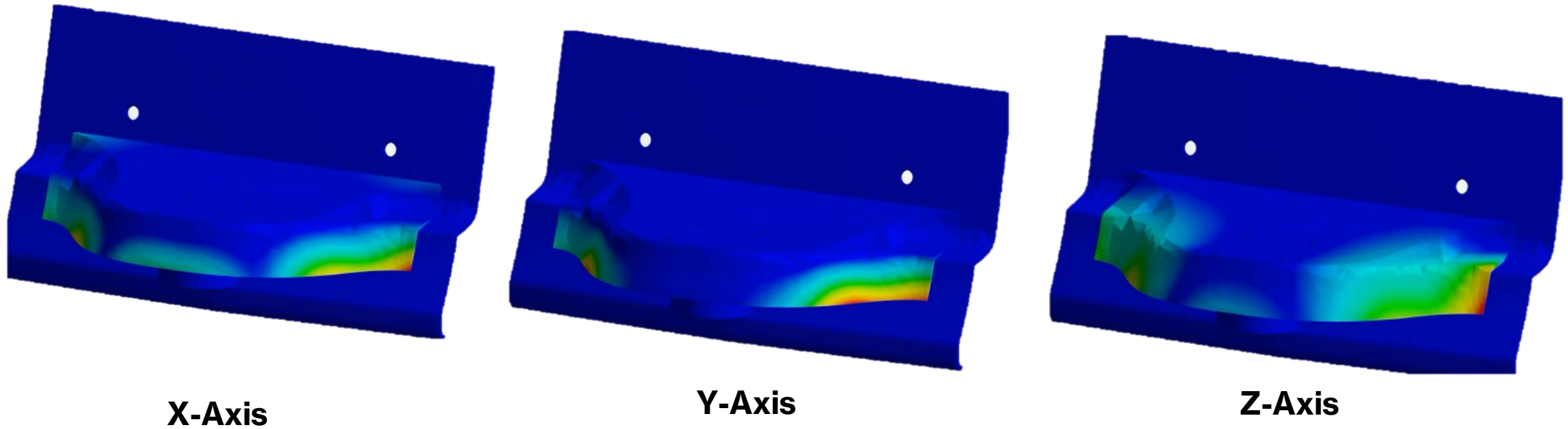
C: Random Vibration
Equivalent Stress
Type: Equivalent Stress
Scale Factor Value: 3 Sigma
Probability: 99.73 %
Unit: MPa
Time: 0 s
2025-11-30 8:00:53 PM



**Bolt mounting
to vehicle**

Rear Mount Design Validation

Random Vibration: Directional Deformation

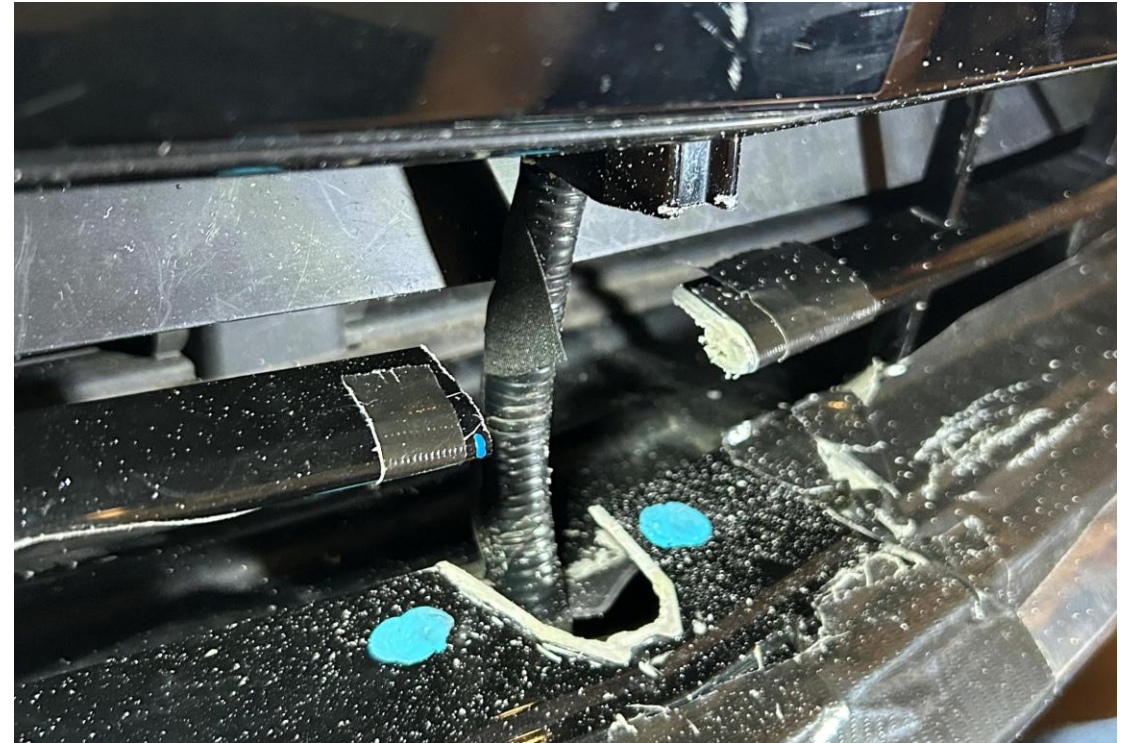


Integration

Front Mount Fabrication



3D Printed

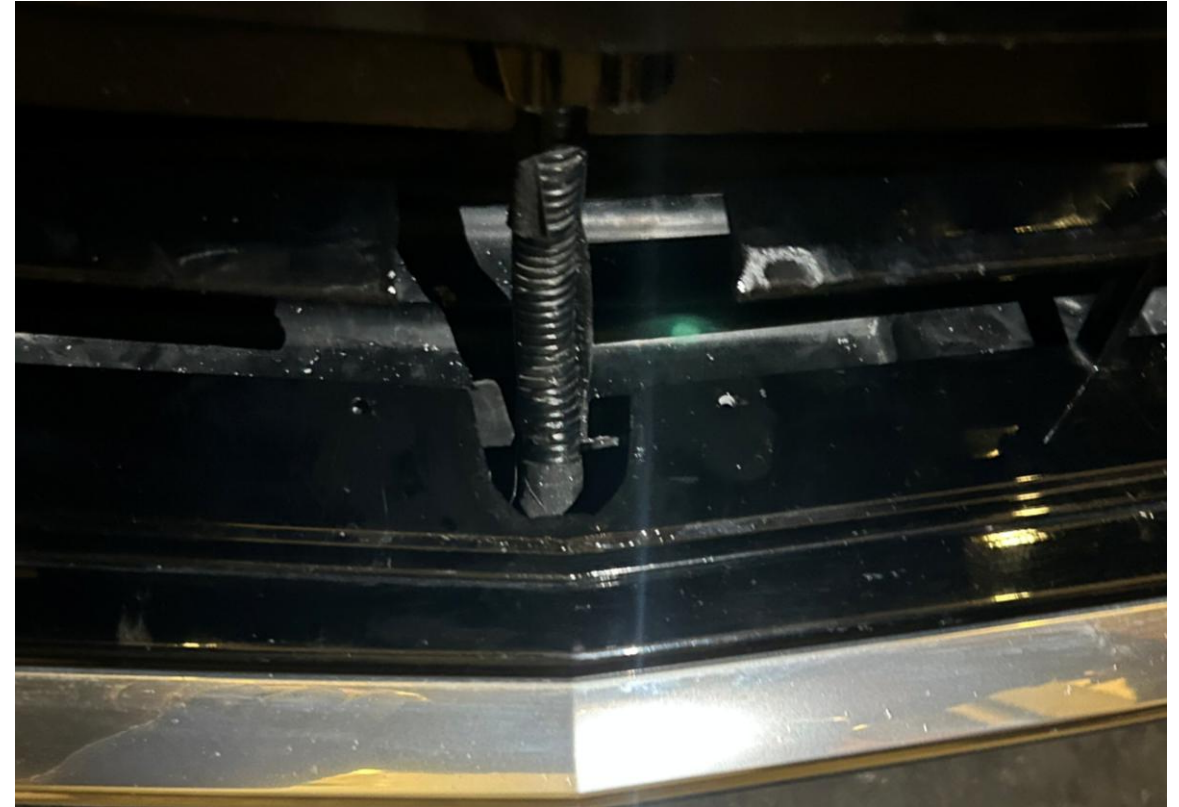


Cut Into Grid Mesh

Front Mount Installation

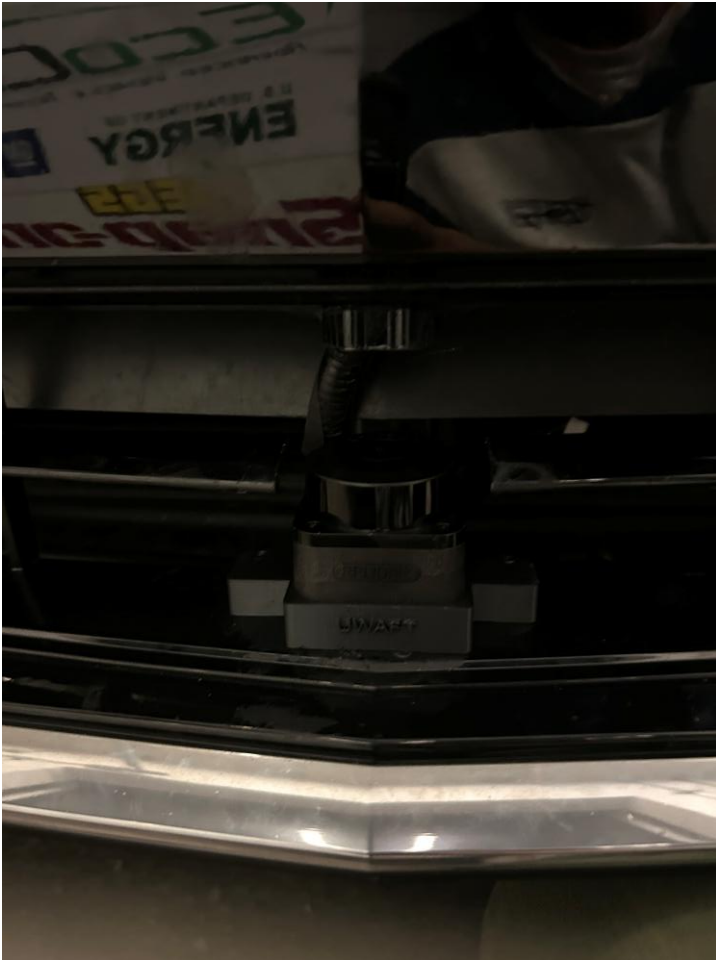


**Mounted & Wire
Repositioning**



**Repainted and Polished for
Aesthetics**

Front Mount Installed!!



Rear Mount Fabrication



3D Printed



Fit Testing

Lessons Learnt

Takeaways from this Project

Lessons Learned

☐ Aesthetics

- ☐ Learned about polishing and finishing plastic surfaces for aesthetics
- ☐ Learned 3D Surface modeling to provide industrial design and integrate mechanical design into a product

☐ Project Management

- ☐ Utilized tools such as JIRA and an Excel Tracker for organization
- ☐ Accessibility is key, for integration

New approaches

☐ UWAFT Guidelines

- ☐ There was ambiguity with project rules such as the 3" envelope, asking for help a lot earlier would have saved the team significant time

☐ Resource Allocations

- ☐ Knowing about all the resources available to us and inquiring about would have helped
- ☐ This includes 3D printing services, simulation software such as ANSYS etc.

Thank You